

Clique-covering co-bridge-free prismatic graphs $\iff$ Coloring bridge-free antiprismatic graphs

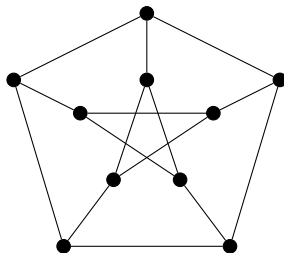
Cléophee Robin
IRIF - Université Paris Cité

Joint work with Eileen Robinson
Université libre de Bruxelles

Février 2026

Structural Graph Theory

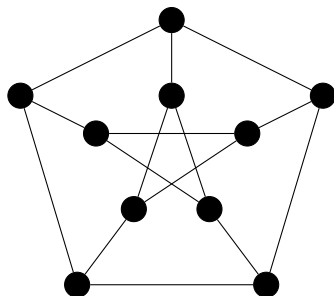
- **Graph** $G = (V, E)$: set of vertices (V) pairwise linked or not by edges E .
- **Optimizations problems** :
 - Shortest path, Minimum spanning tree, Maximum matching...
 - Coloring, Maximum independent set, Hamiltonian cycle...



Restrict the input to some special classes of graphs.

- Do they admit a common structure ?
- are they recognizable in polynomial time ?
- Can we solve NP-complete problem in Polynomial time when restricted to the class ?

Coloring problem



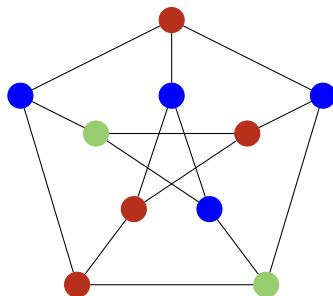
Coloring Problem

INSTANCE: G : a graph

k : a positive integer

QUESTION: Is there a proper coloring of G using at most k colors?

Coloring problem



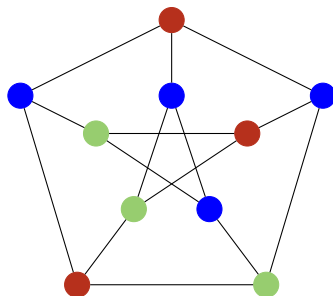
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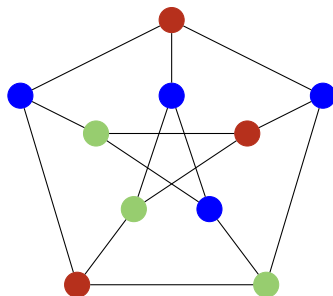


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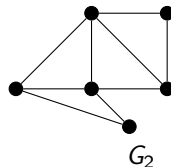
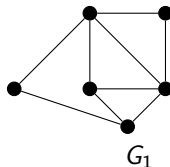
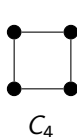
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NP-Complete in the general case

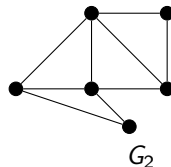
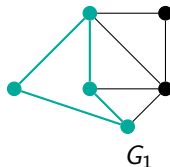
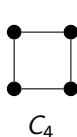
Hereditary classes of graph

- **Induced subgraph of G :** any graph obtained from G by removing some vertices and all their incident edges.



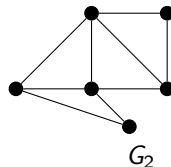
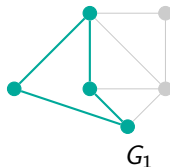
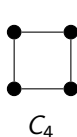
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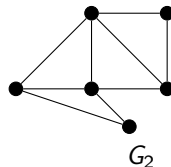
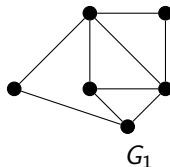
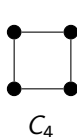
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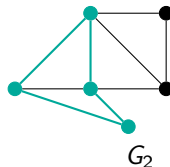
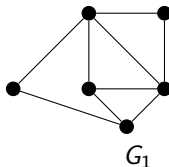
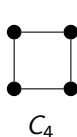
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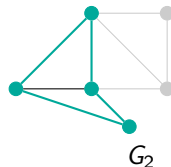
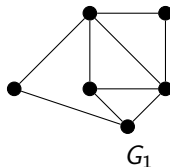
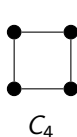
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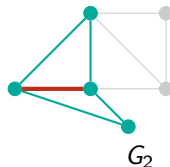
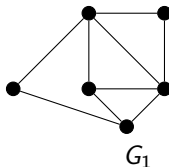
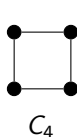
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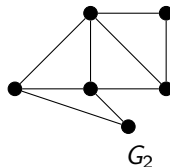
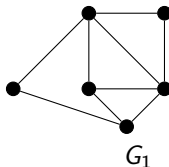
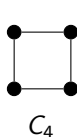
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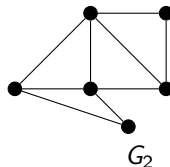
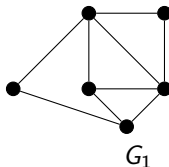
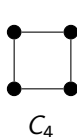
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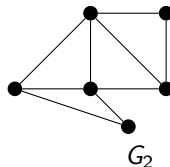
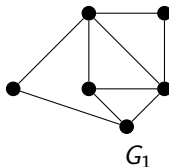
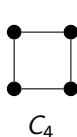
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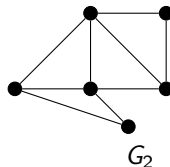
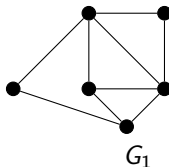
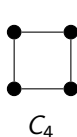
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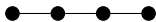


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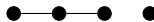
Complexity of the coloring problem

Theorem: Král, Kratochvíl, Tuza and Woeginger [2001]

The Coloring problem is polynomial-time solvable for H -free graphs if H is an induced subgraph of P_4 or of $P_3 + K_1$ and it is NP -complete for any other H .



P_4

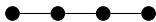


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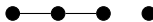
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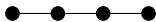
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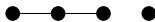
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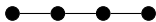
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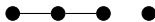
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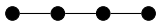
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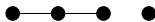
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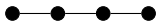
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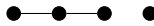
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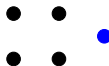
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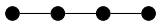
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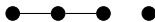
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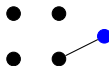
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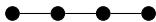
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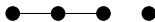
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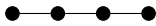
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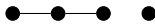
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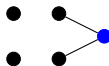
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Property: Lozin and Malyshev

Complexity of Vertex Coloring $\mathcal{F}_2$
 $\Leftrightarrow$ Complexity of Vertex Coloring $\mathcal{F}_3$.

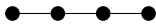
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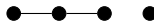
Complexity of the coloring problem

Theorem: Král, Kratochvíl, Tuza and Woeginger [2001]

The Coloring problem is polynomial-time solvable for H -free graphs if H is an induced subgraph of P_4 or of $P_3 + K_1$ and it is NP -complete for any other H .



P_4



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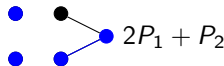
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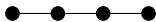
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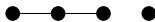
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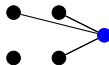
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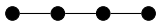
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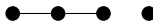
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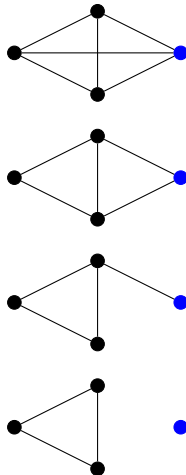
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Prismatic graphs

Definition: Prismatic graphs

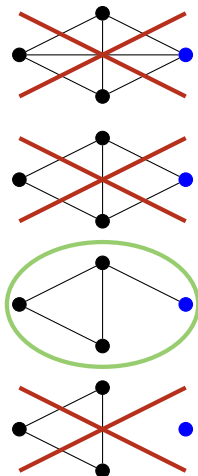
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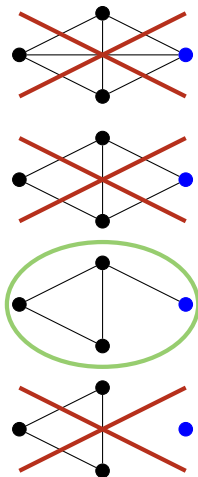


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$\Leftrightarrow$ Complement graphs of $\mathcal{F}_3$

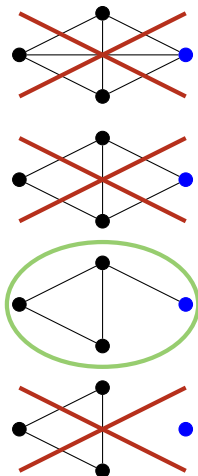
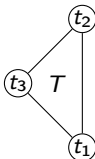
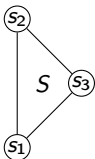


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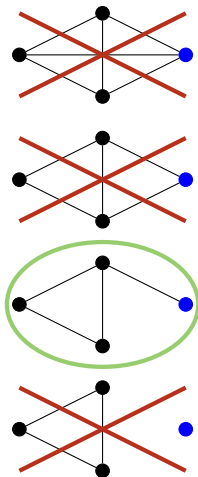
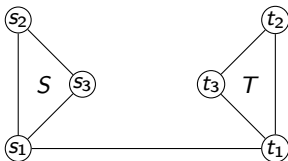


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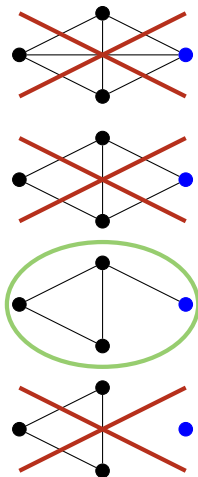
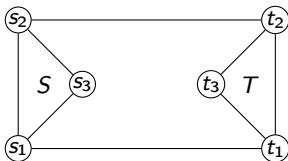


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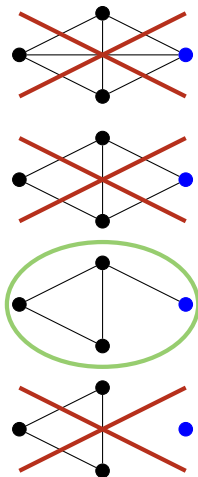
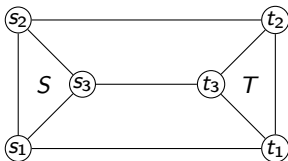


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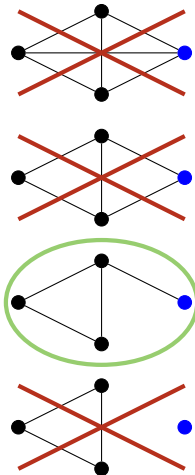
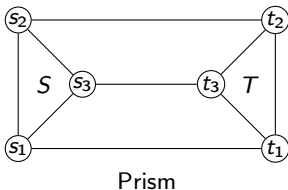


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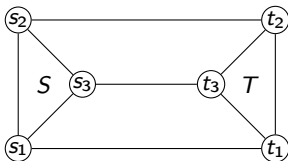


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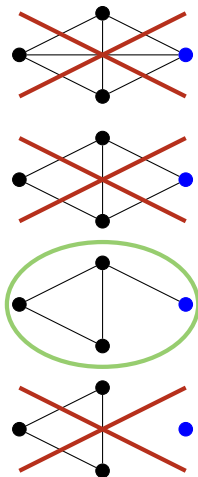
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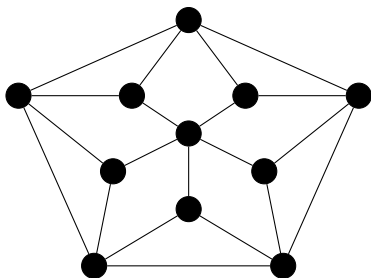


Coloring Problem in $\mathcal{F}_3$

$\Leftrightarrow$ Clique Covering Problem in Prismatic graphs.



Clique Covering Problem

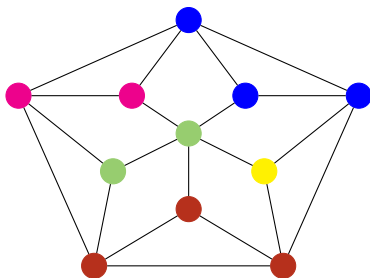


Clique Covering Problem

INSTANCE: G : a graph
 k : a positive integer

QUESTION: Is there a partition of $V(G)$ into at most k cliques?

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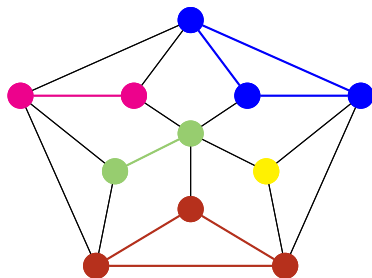


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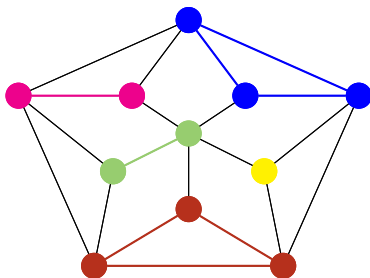


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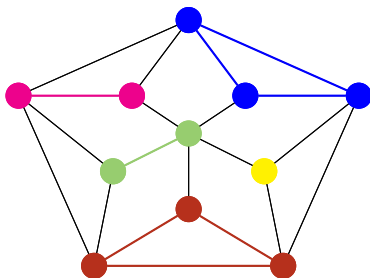
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Coloring graphs in $\mathcal{F}_3 \Leftrightarrow$ Clique covering prismatic graphs.

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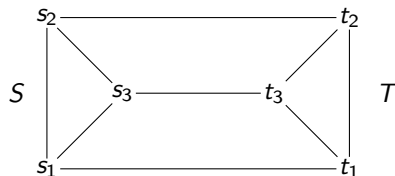
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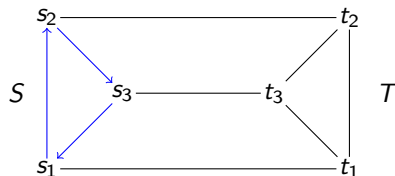
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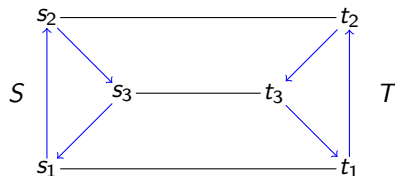
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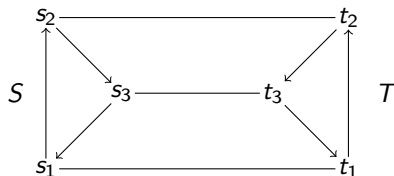
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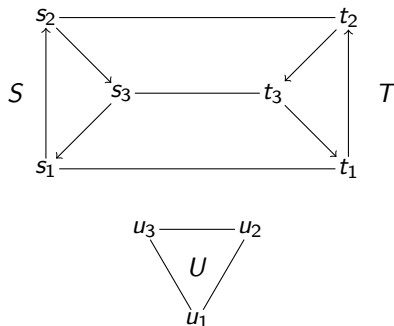
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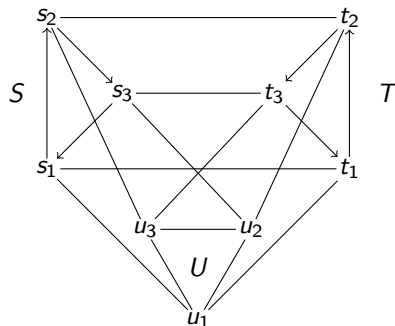
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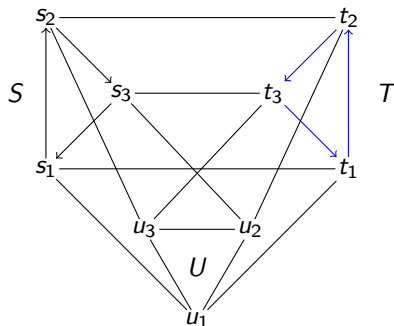
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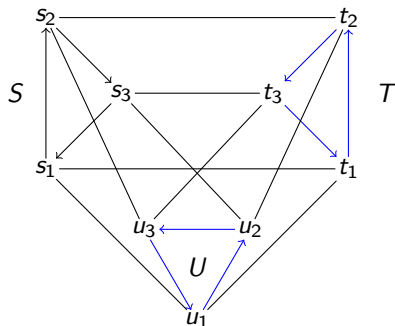
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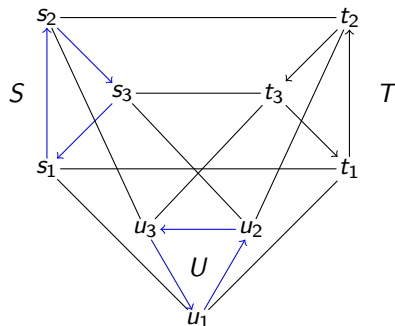
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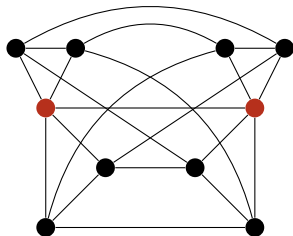
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Definition: Hitting set of triangles

A set S of vertices of G is a **hitting set of triangles** of G if for every triangle T of G $T \cap S \neq \emptyset$.



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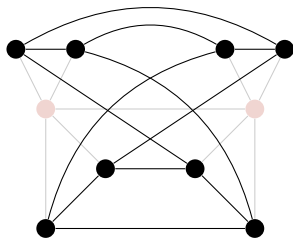
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- Clique covering problem in triangle-free graphs $\Leftrightarrow$ Maximum matching

⇒ If the number of triangles in the optimal partition is known then Clique covering is easy.

Definition: Hitting set of triangles

A set S of vertices of G is a **hitting set of triangles** of G if for every triangle T of G $T \cap S \neq \emptyset$.

Max disjoint triangles $\leq$ Min Hitting set



Hitting set and Clique cover

Theorem [Preissmann, Robin and Trotignon (2021)]

The Clique Covering problem is solvable in $\mathcal{O}(n^{7.5})$ for prismatic graphs with a hitting set of triangles of size 5.

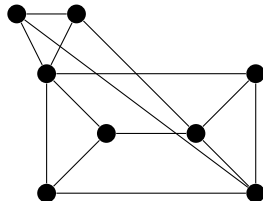
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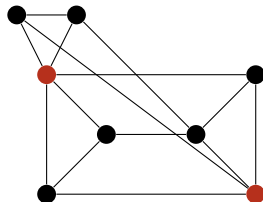
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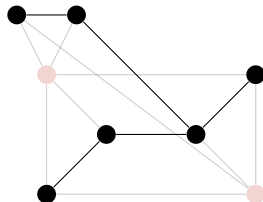
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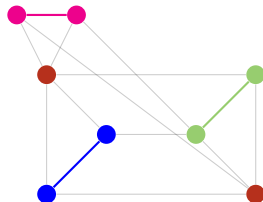
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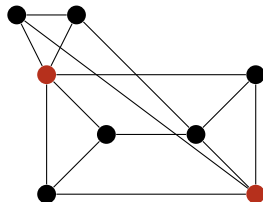
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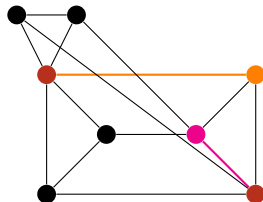
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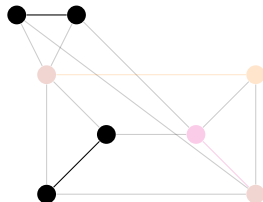
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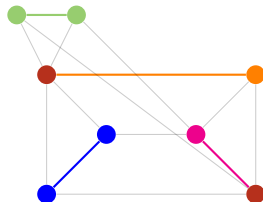
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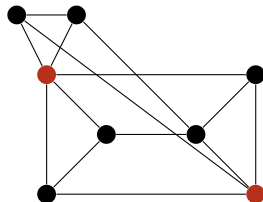
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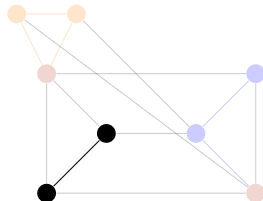
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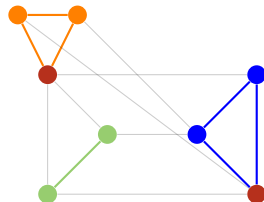
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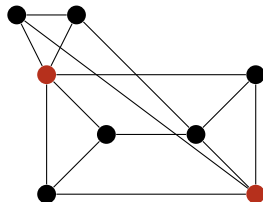
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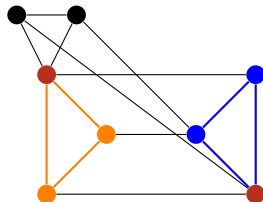
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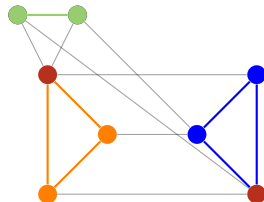
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Non-orientable Prismatic graphs

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Theorem [Preissmann, Robin and Trotignon (2021)]

Every non-orientable prismatic graph either:

- admits a hitting set of triangles of cardinality at most 5
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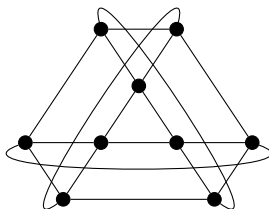
The Clique-covering problem is solvable in $\mathcal{O}(n^{7.5})$ for non-orientable prismatic graphs.

Why doesn't it work in the orientable case?

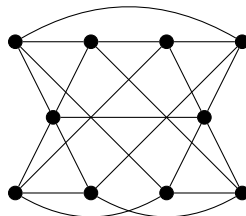
Obstruction for not being orientable:

Chudnovsky and Seymour (2007)

A prismatic graph is orientable if and only if it is in $\text{Free}\{\text{Rotator}, \text{Twister}\}$.



Rotator

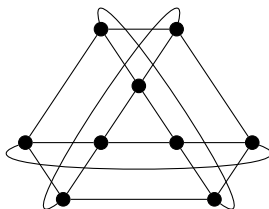


Twister

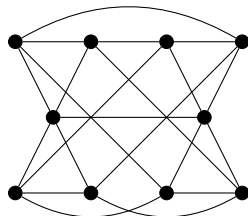
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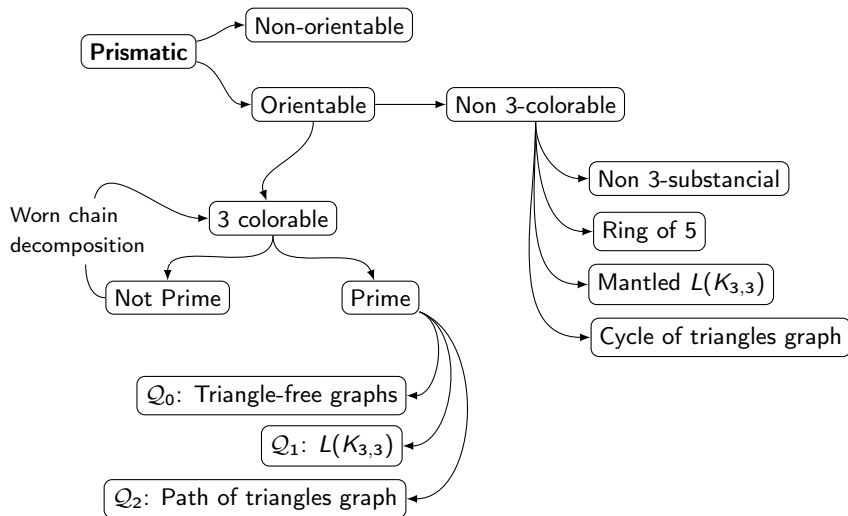
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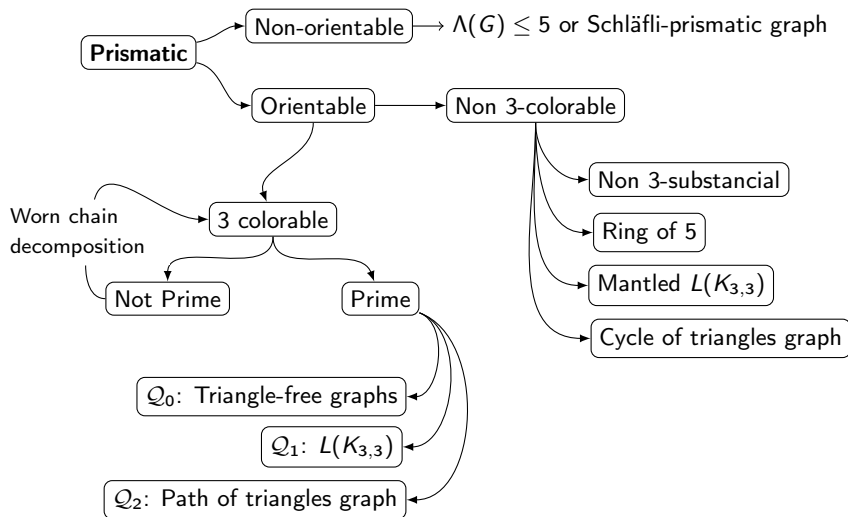
Twister

To have a lot of disjoint triangles, organize the triangle $\Rightarrow$ be orientable.

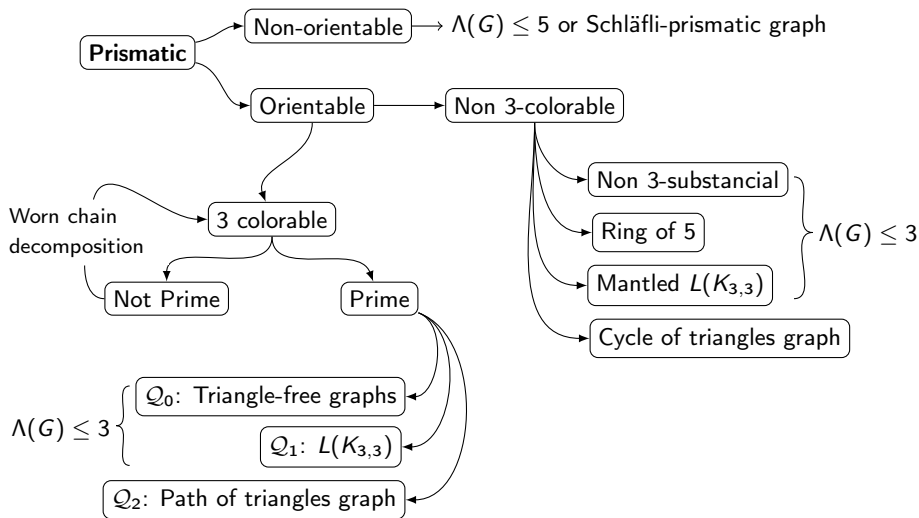
Structure of prismatic graph



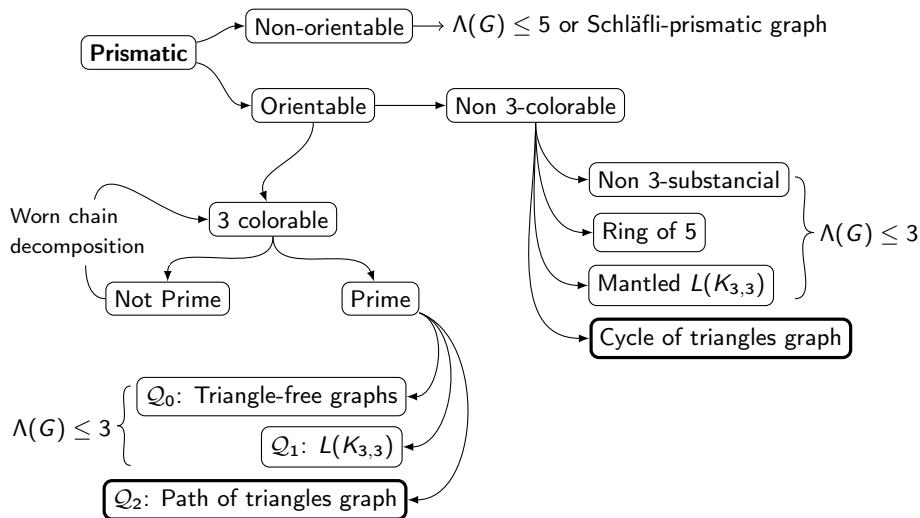
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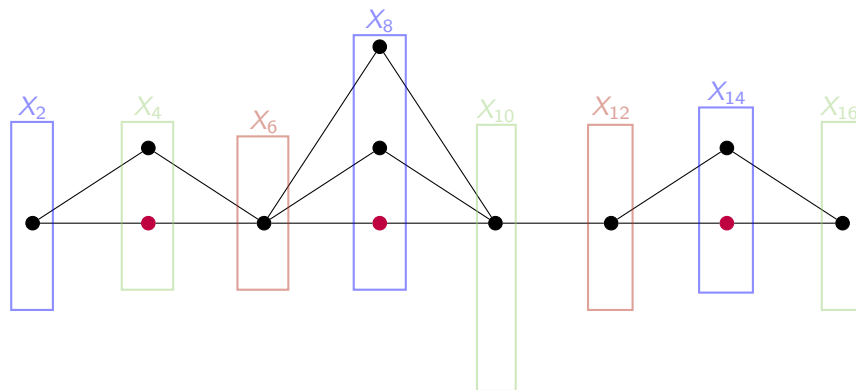
Path of triangles graphs



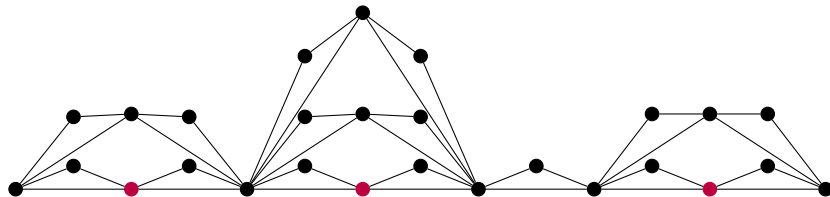
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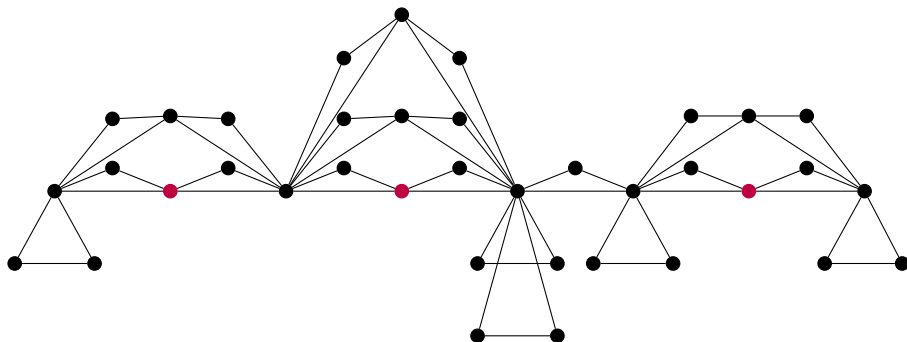
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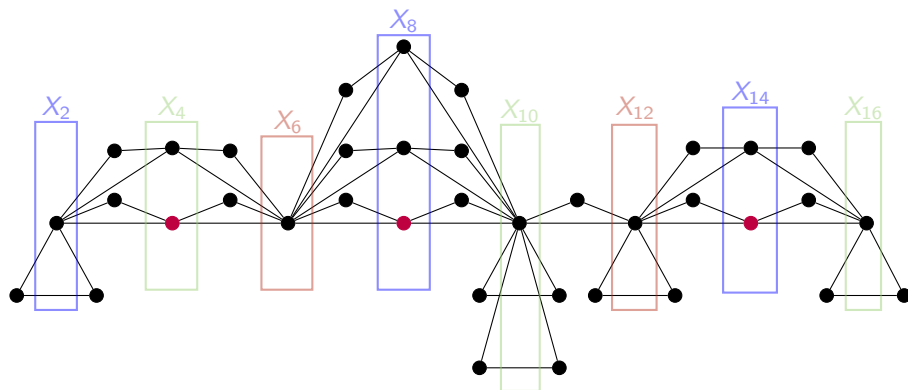
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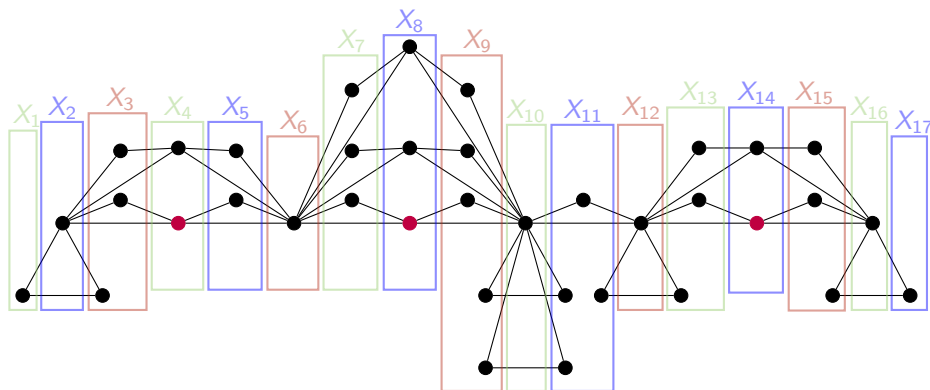
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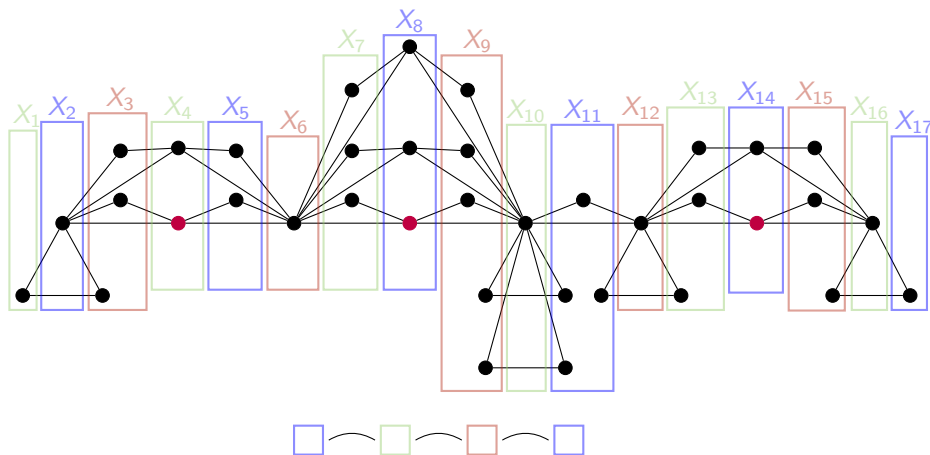
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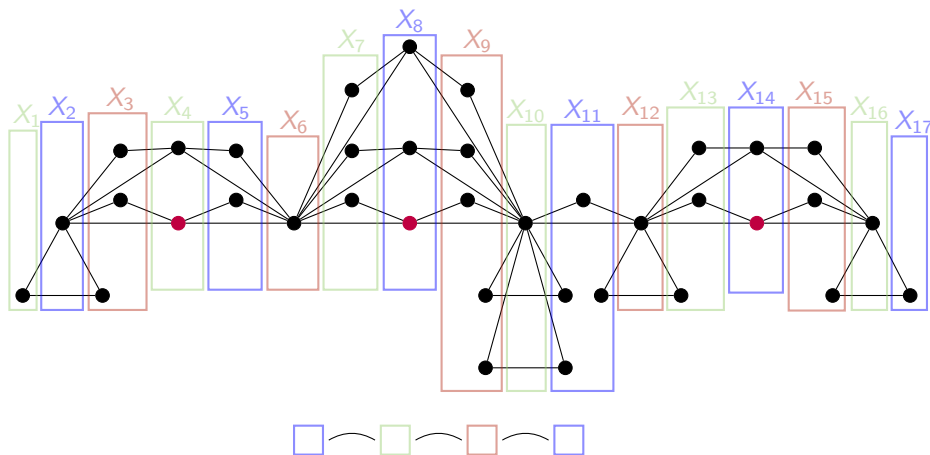
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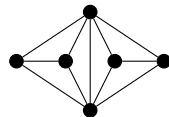
Cycle of triangle graphs = start with cycle of length $3k + 2$.

Hitting set for co-bridge-free prismatic graphs

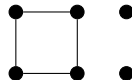
Theorem [Robin and Robinson (2025)]

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Bridge



Co-bridge

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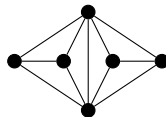
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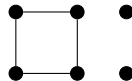
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Ideas for the proof

- Paths and Cycles of triangles graphs can not be too long.
- It is not possible to add too many triangles with the worn chain decomposition.



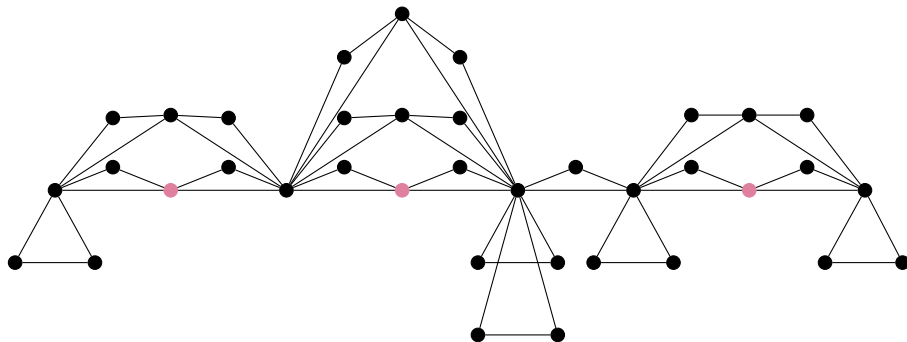
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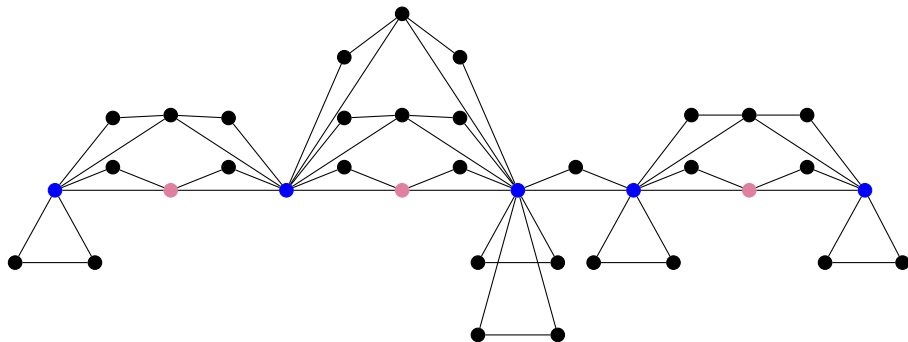
Hitting set for Path of triangles graphs

Those are the only triangles in a path of triangles graphs.



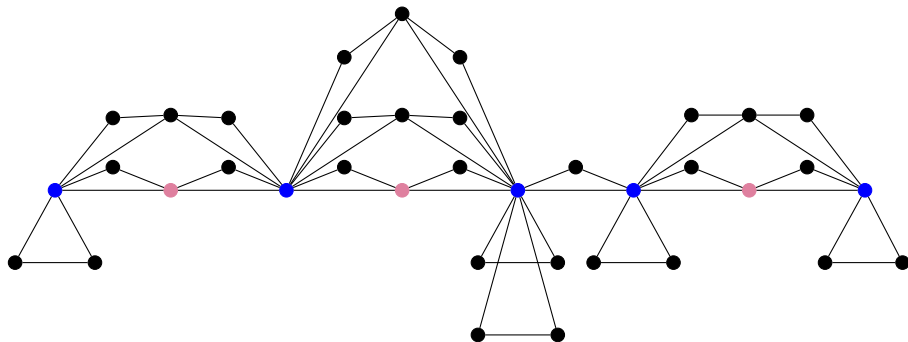
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$| \text{Hitting set} | \leq \text{size of the path.}$

Length of a co-bridge free Path of triangles graphs

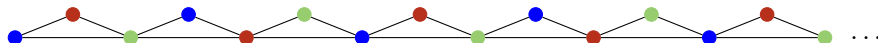
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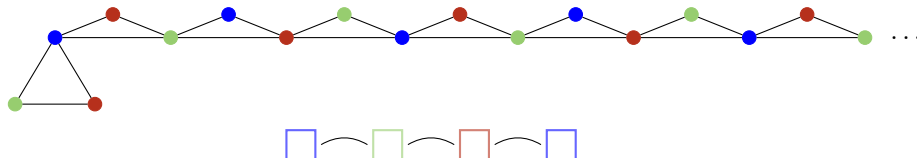
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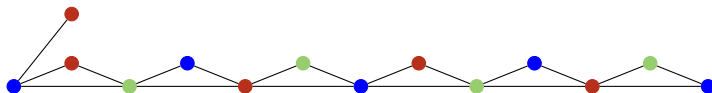
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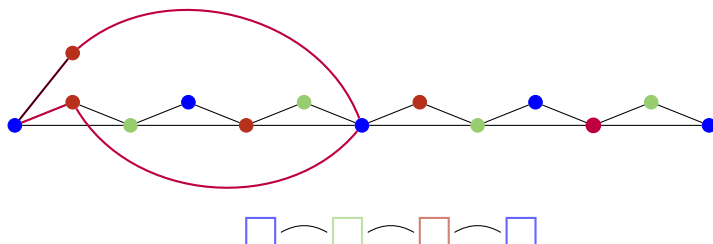
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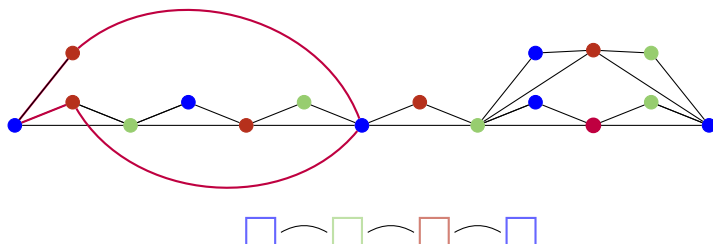
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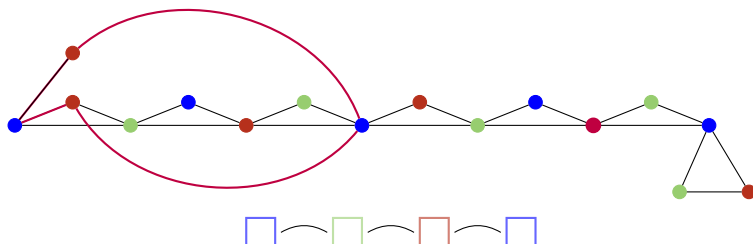
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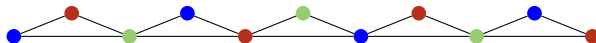
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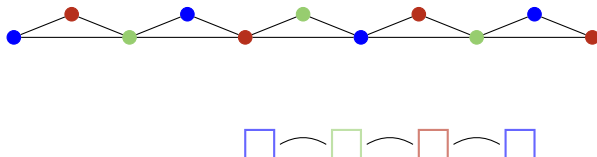
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Lemma

Length at most 6 $\Leftrightarrow$ Hitting set of size at most 6.

Hitting set of a co-bridge free Path of triangles graphs

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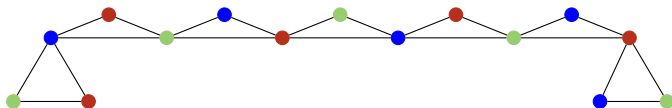


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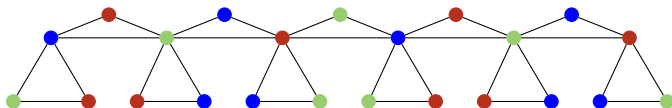


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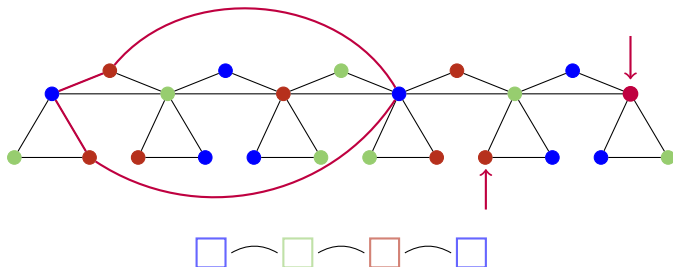


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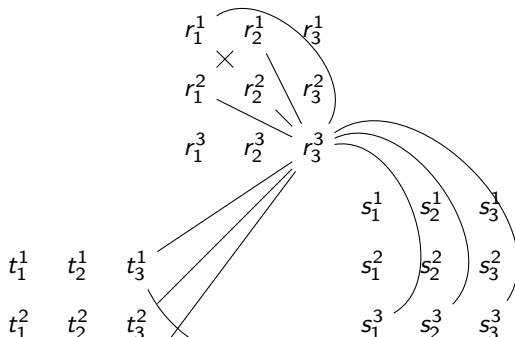
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Thank you!